Latent Environment Navigation for CT-to-MRI Translation via Reinforcement Learning

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[FM Reference] [Experiments (§6)] [Ablation (§6.4)] [RL Methods (§6.5)] [Analysis (§7)]

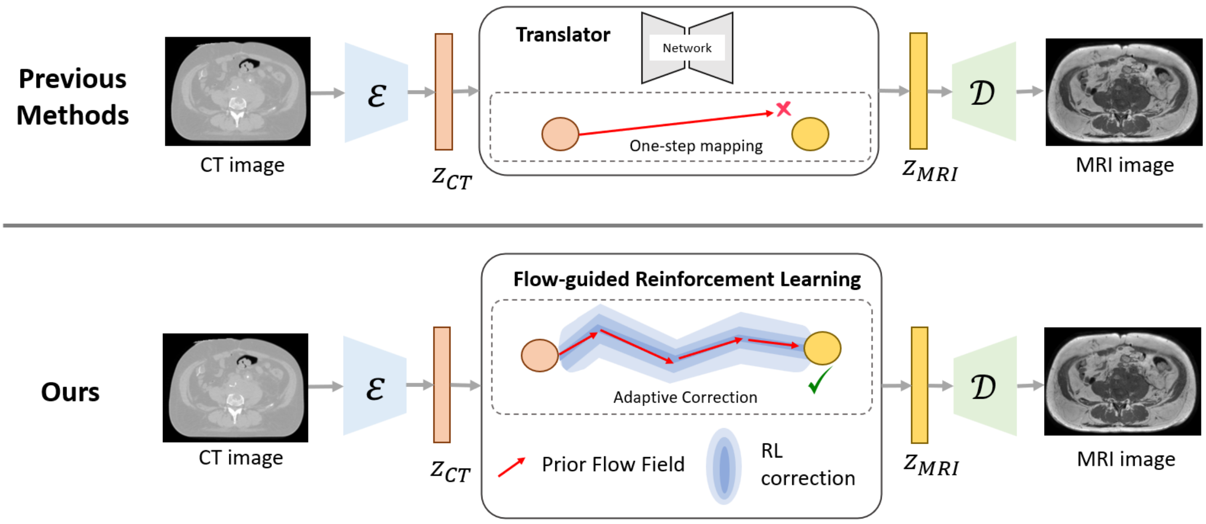


Figure 1: **Overview.** Conventional one-step latent mapping methods (top) rely on a translator network to learn the entire CT-to-MRI transformation in a single forward prediction. As the full modality gap must be bridged at once, these approaches are highly data-hungry and often struggle under limited paired supervision. Moreover, once an inaccurate latent prediction is produced, there is no mechanism for subsequent correction or refinement. In contrast, our method (bottom) reformulates CT-to-MRI translation as a trajectory-guided latent transport process. A Prior Flow Field provides globally consistent transport directions, while a Flow Correction Agent progressively refines the trajectory through adaptive corrections at each transport step.

1 Abstract

CT-to-MRI translation is clinically valuable but remains challenging when only a limited number of paired scans are available — a common constraint in real-world medical settings. Existing latent translation approaches typically formulate the task as a one-step alignment problem, requiring the model to learn a direct cross-modal mapping from sparse supervision.

Such methods are highly sensitive to data scarcity and often suffer from degraded latent correspondence when paired samples are limited.

In this work, we reformulate CT-to-MRI translation as a **multi-step latent transport** problem. Our framework combines a **Flow Matching (FM)** prior, which provides globally consistent transport trajectories in latent space, with an **RL correction policy** that performs adaptive local refinements along the trajectory. To enable stable RL optimization under extremely limited supervision, we introduce the **Flow Improvement Reward (FIR)**: a relative reward that measures improvement over the FM prior rather than absolute distance to the target modality, substantially reducing reliance on large paired datasets.

Experiments demonstrate that the proposed method remains effective with as few as 10 paired scans. Using approximately 1% of the training data required by previous latent alignment methods, our approach achieves SSIM = 0.7300 — surpassing ALDM trained on the full dataset (SSIM = 0.7167). These results suggest that RL-guided latent transport is a promising direction for robust cross-modal medical image translation in low-resource clinical scenarios.

TL;DR We model CT→MRI translation as multi-step latent navigation and use RL to adaptively refine Flow Matching trajectories — achieving better SSIM, PSNR, and MAE than all baselines *with only 10 paired training scans*, addressing the critical data scarcity challenge in medical imaging.

2 Introduction

CT-to-MRI translation is an important task in medical image analysis, enabling MRI-like soft-tissue visualization from widely available CT scans. However, acquiring paired CT–MRI data remains expensive and clinically challenging, making data scarcity a major obstacle for supervised cross-modal translation.

Recent latent-space approaches, such as ALDM, formulate CT-to-MRI translation as a *one-step latent mapping* problem, directly transforming CT latent representations into the target MRI latent space. While effective with sufficient training data, these methods become highly sensitive to limited supervision and often fail to establish reliable cross-modal correspondence when only a small number of paired scans are available. As illustrated in Figure 1, directly learning a single latent transformation can be insufficient for bridging the large modality gap between CT and MRI. This limitation becomes particularly severe under extreme data scarcity, where ALDM collapses when trained with only 10 paired CT-MRI scans (Figure 2).

To address this, we reformulate CT-to-MRI translation as **trajectory-guided latent transport**. Instead of a single-step prediction, our method progressively transports latent representations through 16 intermediate states. A **Flow Matching (FM)** prior provides globally consistent transport directions; an **RL correction policy** applies adaptive local refinements at each step — recovering texture details where FM regresses to the mean. To stabilize learning under extremely limited supervision, we introduce the **Flow Improvement Reward (FIR)**: a relative reward measuring whether each RL step improves over the FM prior, rather than absolute distance to ground truth.

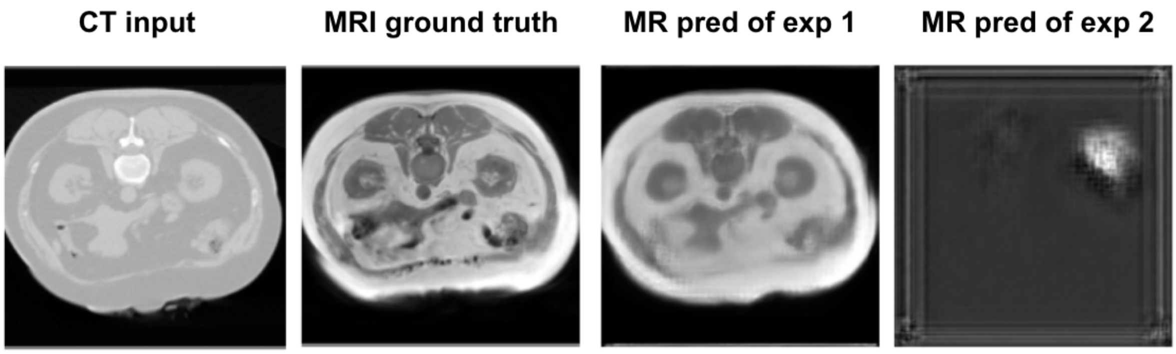


Figure 2: Failure of one-step latent alignment under limited-data training. When trained on the full dataset, ALDM generates anatomically plausible MRI images from CT inputs. However, with only 10 paired CT-MRI scans, the generated outputs collapse into noisy and low-information patterns, indicating that direct latent alignment is highly sensitive to data scarcity.

Main result. Using ∼1% of the training data required by prior latent alignment methods (10 paired scans vs. ∼900 for ALDM), our method achieves SSIM = 0.7300 — surpassing ALDM trained on the full dataset (SSIM = 0.7167).

3 Contributions

- We analyze the failure modes of one-step latent alignment (ALDM) under limited data — showing that VQ-VAE codebook collapse is the root cause, supported by quantitative and latent-space (UMAP/PCA) evidence.
- We propose the first RL framework for latent-space navigation in cross-modal medical image translation, combining a Flow Matching prior (global transport) with an RL correction policy (local texture refinement) — addressing FM's regression-to-mean limitation.
- We introduce the **Flow Improvement Reward (FIR)**, a relative reward signal measuring improvement over the FM prior at each ODE step. FIR enables stable RL training without large paired datasets, as the reward reference comes from FM itself rather than dense ground-truth supervision.
- We demonstrate that our method achieves SSIM = 0.7300 with only 10 paired scans — ∼1% of the training data required by ALDM — surpassing ALDM trained on the full SynthRAD2025 dataset (SSIM = 0.7167, ∼900 scans).

4 Problem Formulation

We formulate CT-to-MRI translation as a latent navigation problem, where a reinforcement learning (RL) agent progressively transports a source CT latent representation toward a target MRI latent representation through a sequence of intermediate states. Instead of learning a direct one-step CT-to-MRI mapping, the agent learns how to navigate the latent space by selecting transport actions that gradually reduce the modality gap.

Given a source CT image x_{CT} and a target MRI image x_{MR}, an encoder maps them into latent representations

$$z_{CT} = E(x_{CT}), \qquad z_{MR} = E(x_{MR}). \tag{1}$$

Starting from $z_0 = z_{CT}$, the policy predicts a continuous transport action a_t at each step, where Δt denotes the integration step size. The latent state then evolves according to

$$z_{t+1} = z_t + \Delta t \cdot a_t, \tag{2}$$

We model latent navigation as a goal-conditioned Markov Decision Process (MDP). The state consists of the current latent representation, the source CT latent code, and the normalized transport time, i.e., $s_t = (z_t, z_{CT}, t)$. At each step, a policy $\pi_\theta(a_t|s_t)$ predicts a transport action that guides the latent trajectory toward the MRI manifold. After T transport steps, the final latent representation is decoded to generate the translated MRI image. The policy is optimized to maximize the cumulative reward

$$R = \sum_{t=0}^{T-1} r_t, \tag{3}$$

where the reward encourages improvement over the Flow Matching prior, high image fidelity, and accurate convergence to the target MRI distribution.

5 Methods

Our proposed *Flow-guided Reinforcement Learning* (FGRL) framework performs CT-to-MRI translation through progressive latent transport within a shared latent space. As illustrated in Figure 3, the framework consists of three key components: (1) a *Prior Flow Generator*, which provides globally consistent transport directions in latent space; (2) a *Latent Transport Environment*, which models the CT-to-MRI transformation as a sequential navigation process; and (3) a *Flow Correction Agent*, which adaptively refines the transport trajectory through reinforcement learning. Together, these components transform CT-to-MRI translation from a one-step latent mapping problem into a trajectory-guided latent navigation process.

5.1 Prior Flow Generator

The Prior Flow Generator is implemented using Flow Matching (FM) [1], which provides a globally consistent transport prior in latent space. Unlike conventional Flow Matching frameworks that learn transport trajectories from random noise to the data distribution, our formulation directly learns a conditional transport process between paired medical modalities. Specifically, the source CT latent representation serves as the starting point of the trajectory, while the target MRI latent representation defines the transport destination. This image-to-image formulation substantially reduces the search space and enables the learned flow field to focus on cross-modal translation rather than unconditional image generation.

Given a source CT latent representation z_{CT} and a target MRI latent representation z_{MR}, a latent point is sampled along the interpolation path

$$z_t = (1-t)z_{CT} + t\, z_{MR} \qquad t \sim \mathcal{U}(0,1). \tag{4}$$

A velocity network v_θ is trained to predict the transport direction from the CT manifold toward the MRI manifold by minimizing

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, z_{CT}, t) - (z_{MR} - z_{CT})\|_2^2 . \tag{5}$$

During inference, the learned velocity field defines a transport trajectory through Euler integration over $N = 16$ steps with integration step size $\Delta t = 1/N$:

$$z_{t+1} = z_t + \Delta t \cdot v_\theta(z_t, z_{CT}, t), \tag{6}$$

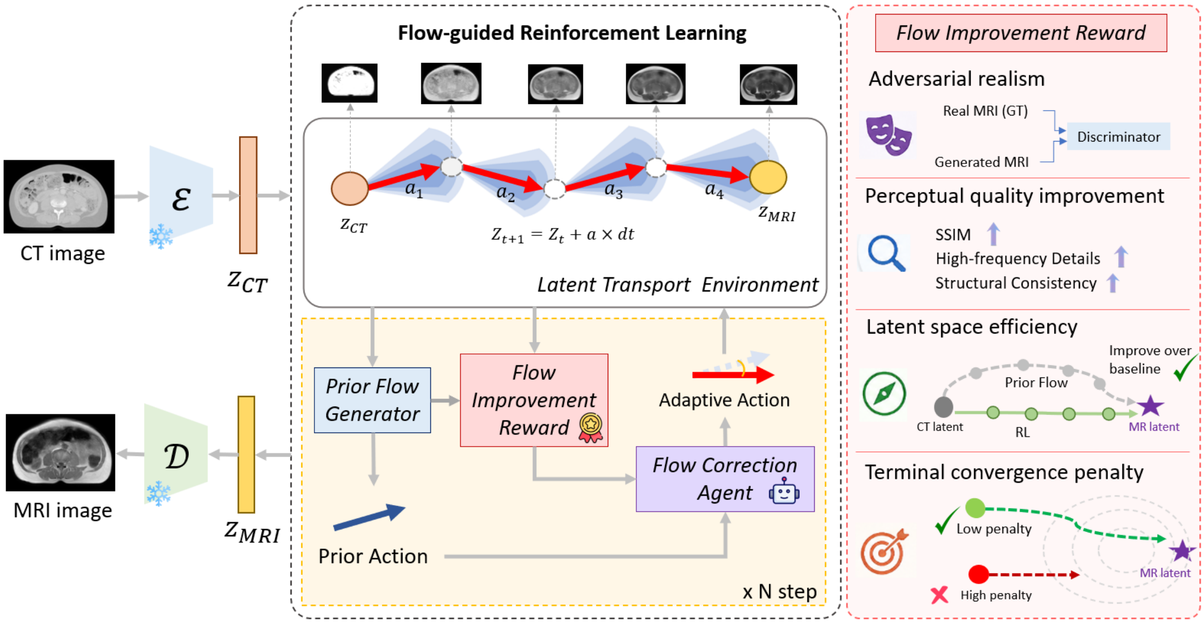


Figure 3: **Flow-guided Reinforcement Learning framework.** The CT image is encoded into latent z_{CT} and progressively transported toward the target MRI latent z_{MRI} within a *Latent Transport Environment*. A *Prior Flow Generator* defines the initial transport direction; a *Flow Correction Agent* predicts adaptive corrections at each integration step. The agent is trained with a *Prior-Flow Improvement Reward* combining: (1) adversarial realism, (2) perceptual fidelity, (3) trajectory improvement over the FM baseline, and (4) terminal convergence to the target MRI manifold.

The resulting trajectory serves as the *Prior Flow* used throughout the latent transport process. Although Flow Matching provides stable global transport, its regression objective often leads to regression-to-the-mean behavior and loss of fine anatomical details under limited-data conditions. To address this limitation, we introduce a Flow Correction Agent that adaptively refines the transport trajectory through reinforcement learning.

5.2 Flow Correction Agent

Rather than learning the entire CT-to-MRI transport process from scratch, the Flow Correction Agent learns adaptive corrections on top of the Prior Flow generated by Flow Matching.

Given the prior flow direction v_{flow}, the agent predicts a raw correction action a_{raw}. To improve controllability, the correction is decomposed into flow-parallel and flow-orthogonal components. The normalized flow direction is first computed as

$$\hat{v}_{\text{flow}} = \frac{v_{\text{flow}}}{|v_{\text{flow}}|_2}. \tag{7}$$

The raw correction is then decomposed into

$$a_{\parallel} = (a_{\text{raw}} \cdot \hat{v}_{\text{flow}})\hat{v}_{\text{flow}}, \tag{8}$$

$$a_{\perp} = a_{\text{raw}} - a_{\parallel}. \tag{9}$$

The coefficients λ_p and λ_o control the influence of parallel and orthogonal corrections, respectively. The final transport action is defined as

$$a_t = v_{\text{flow}} + \lambda_p a_{\parallel} + \lambda_o a_{\perp}, \tag{10}$$

Intuitively, the parallel component adjusts the transport speed along the Prior Flow trajectory, while the orthogonal component enables local manifold corrections that cannot be captured by the Flow Matching model alone. The latent state is then updated according to

$$z_{t+1} = z_t + \Delta t \cdot a_t. \tag{11}$$

5.3 Prior-Flow Improvement Reward

A key challenge in applying reinforcement learning to CT-to-MRI translation is reward design. Directly optimizing reconstruction quality often results in unstable learning, particularly under limited paired supervision. Instead of learning the entire cross-modal transformation from scratch, we use the Prior Flow trajectory generated by Flow Matching as a reference and reward the agent only when it improves upon this baseline.

Let z_{flow} denote the latent representation generated by the Prior Flow Generator and z_{RL} denote the latent representation obtained after applying the Flow Correction Agent. The proposed Prior-Flow Improvement Reward consists of four complementary components.

Adversarial Realism. Flow Matching tends to produce overly smooth predictions due to its regression objective. To encourage the generated MRI images to lie on the target MRI distribution, we employ a frozen discriminator and define

$$r_{\text{adv}} = \sigma(D(z_{\text{RL}}))\,, \tag{12}$$

where $D(\cdot)$ denotes the discriminator and $\sigma(\cdot)$ is the sigmoid function. Higher values indicate more realistic MRI representations.

Perceptual Quality Improvement. To recover anatomical structures that may be lost during transport, we reward improvements in image quality relative to the Prior Flow baseline. Specifically, the structural similarity improvement is defined as

$$r_{\text{SSIM}} = \text{SSIM}(x_{\text{RL}}, x_{MR}) - \text{SSIM}(x_{\text{flow}}, x_{MR}). \tag{13}$$

where x_{RL} and x_{flow} are the decoded MRI images generated from z_{RL} and z_{flow}, respectively.

To further encourage recovery of fine anatomical structures, we reward improvements in high-frequency image details. Let $\text{HF}(\cdot)$ denote a high-frequency reconstruction error metric. The corresponding reward is defined as

$$r_{\text{HF}} = \text{HF}(x_{\text{flow}}, x_{MR}) - \text{HF}(x_{\text{RL}}, x_{MR}). \tag{14}$$

Latent Transport Improvement. Rather than optimizing absolute reconstruction quality, the proposed reward evaluates whether the RL trajectory improves upon the Prior Flow trajectory in latent space. The latent improvement reward is defined as

$$r_{\text{latent}} = \|z_{\text{flow}} - z_{MR}\|_1 - \|z_{\text{RL}} - z_{MR}\|_1. \tag{15}$$

A positive reward is obtained when the corrected trajectory reduces the latent transport error compared with the Prior Flow baseline.

Terminal Convergence Penalty. Although dense rewards improve local trajectory quality, they do not guarantee accurate convergence to the target MRI manifold. Therefore, a terminal penalty is applied at the final transport step:

$$r_{\text{term}} = -\|z_{\text{RL}} - z_{MR}\|_1. \tag{16}$$

This term encourages the trajectory to terminate near the target MRI latent representation while still allowing exploration during earlier transport steps.

Overall Reward. The final Prior-Flow Improvement Reward is computed as

$$r_t = \lambda_{\text{latent}} r_{\text{latent}} + \lambda_{\text{SSIM}} r_{\text{SSIM}} + \lambda_{\text{HF}} r_{\text{HF}} + \lambda_{\text{adv}} r_{\text{adv}} + \lambda_{\text{term}} r_{\text{term}}, \tag{17}$$

where the coefficients balance latent transport accuracy, perceptual fidelity, adversarial realism, and terminal convergence. By measuring improvement relative to the Prior Flow trajectory, the reward encourages the Flow Correction Agent to refine transport paths instead of rediscovering the entire CT-to-MRI mapping from scratch.

5.4 Policy Optimization

We investigate two policy optimization strategies corresponding to the two versions of the proposed Flow-guided Reinforcement Learning (FGRL) framework. FGRL-Lite employs REINFORCE as a lightweight proof-of-concept implementation, whereas FGRL-Full adopts Proximal Policy Optimization (PPO) to improve optimization stability and sample efficiency. Both versions operate in the same CT-to-MRI latent transport environment and use the same training dataset.

FGRL-Lite: REINFORCE. For the lightweight version of the framework, the Flow Correction Agent is optimized using REINFORCE with an actor–critic formulation. The actor is implemented as a UNet policy network (channels 32–64–64), while the critic estimates the state value function. The actor objective is

$$\mathcal{L}_{\text{actor}} = -\mathbb{E}\left[\log \pi_\theta(a_t \mid s_t)\, A_t\right]. \tag{18}$$

and the critic is trained using

$$\mathcal{L}_{\text{critic}} = \left\| V_\phi(s_t) - R_t \right\|_2^2. \tag{19}$$

The overall objective is

$$\mathcal{L} = \mathcal{L}_{\text{actor}} + \lambda_v \mathcal{L}_{\text{critic}} - \lambda_e \mathcal{H}(\pi_\theta), \tag{20}$$

where $\mathcal{H}(\pi_\theta)$ denotes the policy entropy. The advantage estimate is

$$A_t = R_t - V_\phi(s_t), \tag{21}$$

and the Monte Carlo return is

$$R_t = \sum_{k=t}^{T} \gamma^{k-t} r_k, \tag{22}$$

with discount factor $\gamma = 0.99$. Entropy regularization encourages exploration during training.

FGRL-Full: PPO. To improve optimization stability and sample efficiency, the full version of the framework replaces REINFORCE with Proximal Policy Optimization (PPO). PPO constrains policy updates through a clipped surrogate objective

$$\mathcal{L}_{\text{clip}} = \mathbb{E}\left[\min\big(r_t(\theta)A_t, , \text{clip}\big(r_t(\theta), 1-\varepsilon, 1+\varepsilon\big)A_t\big)\right], \tag{23}$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \tag{24}$$

is the probability ratio between the current and previous policies.

The PPO objective is

$$\mathcal{L}_{\text{PPO}} = \mathcal{L}_{\text{clip}} + \lambda_v \left\| V_\phi(s_t) - R_t \right\|^2 - \lambda_e \mathcal{H}(\pi_\theta). \tag{25}$$

In our implementation, PPO uses a clipping threshold $\varepsilon = 0.1$ and performs four policy-update epochs per rollout. PPO is adopted in Phase 2 because it provides more stable optimization for long-horizon latent transport trajectories while preserving the exploration capability required for trajectory refinement.

6 Experiments

Framework versions. All experiments are conducted under the same limited-data setting using only 10 paired CT–MRI scans. We evaluate two versions of the proposed Flow-guided Reinforcement Learning (FGRL) framework.

FGRL-Lite. A proof-of-concept implementation consisting of a lightweight Flow Matching prior and a simple reinforcement learning agent optimized with REINFORCE. The flow model is implemented using a shallow residual convolutional architecture trained with the standard velocity-matching objective. This version serves as a minimal latent-transport baseline and is used to verify that RL-guided trajectory refinement improves upon both pure Flow Matching and one-step latent alignment methods.

FGRL-Full. The complete version of the proposed framework, which enhances both the Prior Flow Generator and the Flow Correction Agent. The Flow Matching prior is upgraded to an attention-enhanced UNet architecture with multi-scale feature aggregation and high-frequency preservation objectives, producing more accurate transport trajectories in latent space. In addition, REINFORCE is replaced with PPO and the full Prior-Flow Improvement Reward is employed for policy optimization. This version represents the final Flow-guided Reinforcement Learning framework evaluated throughout the paper.

6.1 Dataset

We use the **SynthRAD2025 Grand Challenge** dataset, which contains approximately 900 paired CT–MRI scans across multiple anatomical regions. All scans are spatially aligned and suitable for supervised cross-modality image translation.

Following the objective of studying CT-to-MRI translation under extreme data scarcity, all experiments are conducted using only **10 paired abdomen CT–MRI scans**. This setting represents approximately 1% of the available training data and is intentionally designed to evaluate whether the proposed Flow-guided Reinforcement Learning framework can learn meaningful latent transport trajectories with minimal paired supervision.

6.2 Experimental Setup

All methods (except ALDM) operate in a shared latent space established by two modality-specific VAEs. Both CT and MRI VAEs use the AutoencoderKL architecture with three resolution levels, channel multipliers $[1, 2, 4]$, 128 base channels, and two residual blocks per level. The latent representation has size $3 \times 64 \times 64$, and the KL regularization weight is set to 10^{-6}.

To evaluate the impact of model design, we consider two versions of the proposed framework. **FGRL-Lite** employs a lightweight Flow Matching prior based on a residual convolutional network trained using only the standard velocity-matching loss. **FGRL-Full** replaces this backbone with an attention-enhanced UNet Flow Matching model and introduces additional high-frequency preservation objectives during training.

For policy learning, FGRL-Lite uses REINFORCE, whereas FGRL-Full adopts PPO together with the complete Prior-Flow Improvement Reward. Both versions use the same latent transport environment, trajectory length ($N = 16$ transport steps), and training dataset.

6.3 Main Results

Table 1: Comparison of different latent translation paradigms under the limited-data setting (10 paired CT–MRI scans). The last row reports ALDM trained on the full SynthRAD2025 dataset as a reference. Best results are shown in **bold green**, and the overall best result is additionally underlined.

Method	Paradigm	SSIM↑	PSNR↑
ALDM	One-step latent alignment	0.1023	7.486
Flow Matching (Lite)	Multi-step latent transport	0.4916	16.4415
FGRL-Lite	Multi-step transport + RL	**0.5107**	**16.6080**
Flow Matching (Full)	Multi-step latent transport	0.7233	20.8050
FGRL-Full	Multi-step transport + RL	**0.7300**	**20.9903**
ALDM (Full data)	One-step latent alignment	0.7167	16.5200

The progression 1-step → multi-step → RL-guided directly mirrors our hypothesis: more steps and per-step adaptivity yield better synthesis under limited data.

Data Efficiency Highlight

Our method uses **10 paired scans**. ALDM on the full SynthRAD2025 dataset (∼900 scans) achieves SSIM = 0.7167. Our Flow-guided RL with ∼90× fewer scans reaches **SSIM** = 0.7300.

ALDM (full dataset) < **Ours (10 scans)**

0.7167 0.7300

SSIM · ∼900 scans SSIM · 10 scans

Note: All methods are evaluated on the same test set. The comparison highlights the data efficiency of the proposed method under extremely limited paired training data.

6.4 Ablation Study

A — Is the Flow Matching Prior Necessary?

We compare *Raw RL* (learns transport from scratch) against *Flow-guided RL* (steers an FM trajectory). Both use REINFORCE on the 10-scan setting.

Table 2: Effect of Flow Matching prior (10-scan REINFORCE).

Variant	FM Prior	SSIM ↑	PSNR ↑	MAE ↓
Raw RL (w/o image reward)	×	0.3558	12.430	0.1987
Raw RL (with image reward)	×	0.3601	12.9503	0.1914
FGRL-Lite	✓	**0.5107**	**16.6080**	**0.0412**

Raw RL (0.3558–0.3601) performs *worse* than FM alone (0.4916). Without a transport prior, the policy must simultaneously discover direction, magnitude, and structure of latent transport — an over-specified problem under limited data. The FM prior is not merely helpful; it is **necessary**.

B — Where Should RL Intervene in the ODE?

We compare two RL placement strategies in the PPO setting. *Post-flow* acts on the FM endpoint z_{flow}; *interleaved* acts at each of the 16 ODE steps.

$$\text{Interleaved:}\quad z_{t+1} = z_t + \big(v_{FM}(z_t, t) + \delta v_{RL}(z_t, z_{CT}, t)\big)\Delta t \tag{26}$$

$$\text{Dense reward}_t = \big\|z_t^{FM} - z_{MR}\big\|_1 - \big\|z_t^{RL} - z_{MR}\big\|_1 \tag{27}$$

Table 3: RL placement ablation (PPO, body-mask SSIM).

Variant	RL acts at	dist(z, z_{MR})	SSIM	Δ vs FM
Flow Matching only	—	—	0.7233	—
Post-flow PPO	FM endpoint	≈ 0.09 (flat)	0.7246	+0.0013
Interleaved PPO (FGRL-Full)	Each of 16 ODE steps	≈ 1.09 (rich)	**0.7300**	**+0.0067**

Interleaved placement achieves ∼**9× the gain** of post-flow. At the FM endpoint, dist$(z_{\text{flow}}, z_{MR}) \approx 0.09$ — the reward landscape is nearly flat and the policy receives no useful gradient. Starting from z_{CT} (dist ≈ 1.09) gives 16 steps of non-zero reward signal per episode.

6.5 RL Algorithm Comparison

We explored three algorithms across the two phases. This is a developmental narrative, not a controlled comparison — settings differ between REINFORCE and PPO/SAC.

PPO results (body-mask SSIM):

*Interleaved PPO with terminal-only rewards improves both SSIM and PSNR over the FM baseline — unlike post-flow correction, which operates on a near-flat reward landscape (*dist ≈ 0.09*) and gains only* +0.0007.

Table 4: RL algorithms explored across phases.

Algorithm	Description	Outcome
REINFORCE	Vanilla policy gradient with advantage baseline.	SSIM 0.5107 (+0.02 over FM-Lite). Stable at toy experiment.
SAC	Off-policy actor-critic with entropy maximization.	Complete failure — Q-function collapses to 0.
PPO	On-policy clipped surrogate (ε=0.1, 4 epochs/rollout).	SSIM 0.7300 (+0.0067 over FM). Best with action_scale 0.1.

SAC Q-collapse. In the post-flow setting, $\mathrm{dist}(z_{\mathrm{flow}}, z_{MR}) \approx 0.09$. Reward variance across actions falls below 10^{-3}; the Bellman target collapses to $Q(s, a) \approx 0$; the actor receives no gradient. Off-policy replay amplifies the problem: stale near-zero-reward transitions dominate.

Key Insight

Algorithm choice matters less than **where** RL intervenes. Both REINFORCE and PPO improve FM when applied *interleaved*. SAC fails not because of the algorithm, but because the post-flow reward landscape is too flat for *any* RL algorithm to learn from.

6.6 Visualization

We further provide qualitative and latent-space visualizations to better understand the behavior of the proposed Flow-guided Reinforcement Learning (FGRL) framework.

Figure 4 shows representative CT-to-MRI translation results generated by the complete FGRL framework. Even under the extremely limited-data setting, the model produces MRI images that are visually close to the ground truth while preserving the global anatomical layout of the input CT image.

Figure 5 illustrates the intermediate predictions throughout the 16-step transport process. Rather than performing an abrupt one-step translation, the latent representation evolves gradually toward the MRI manifold, demonstrating that the proposed framework learns a continuous transport trajectory in latent space.

Figure 6 compares the outputs generated by the Prior Flow Generator and the proposed FGRL-Full model. Since the Flow Matching prior already provides a strong globally consistent transport trajectory, the improvements introduced by RL are relatively small and may be difficult to perceive visually. Nevertheless, the numerical metrics reported below each sample consistently demonstrate improvements in reconstruction quality, indicating that RL performs effective sample-specific trajectory refinement.

Figure 7 highlights the effect of reinforcement learning when the underlying Flow Matching prior is relatively weak. Compared with the lightweight Flow Matching baseline, FGRL-Lite produces noticeably sharper anatomical structures and better contrast consistency. The improvement is substantially more visible than in the FGRL-Full case, illustrating that reinforcement learning

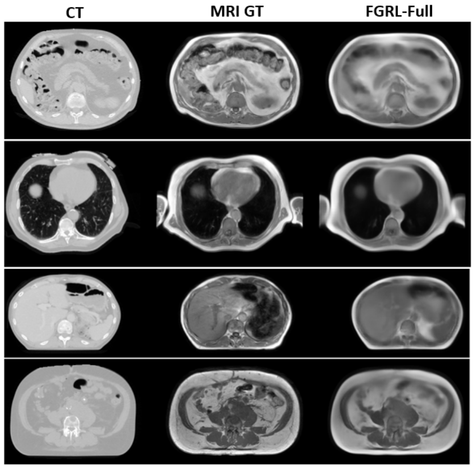


Figure 4: Qualitative CT-to-MRI translation results produced by **FGRL-Full**. From left to right: input CT image, ground-truth MRI, and the generated MRI. The proposed method successfully preserves large anatomical structures while recovering realistic MRI contrast and fine tissue boundaries.

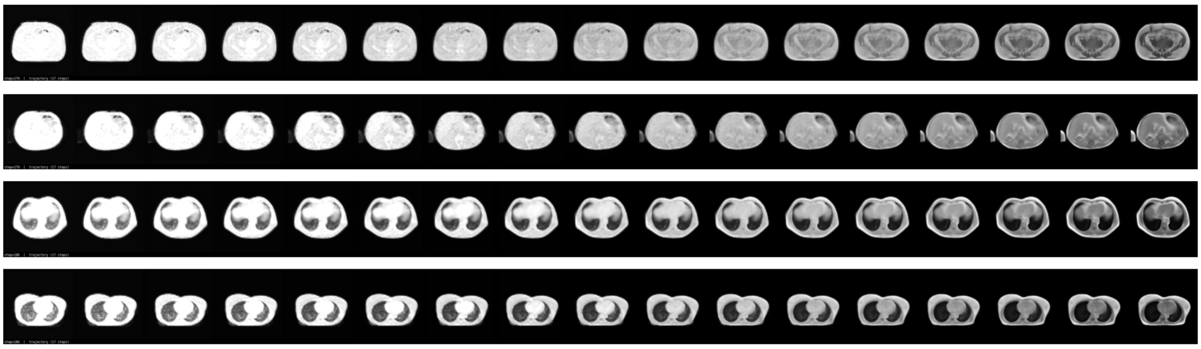


Figure 5: Visualization of the 16-step latent transport process. Starting from the input CT latent representation, the generated image is progressively refined toward the target MRI appearance through sequential latent transport updates.

mainly acts as a local correction mechanism that complements the global transport direction learned by Flow Matching.

7 Analysis and Discussion

7.1 Why Multi-Step Beats One-Step Under Limited Data

ALDM's VQ-VAE codebook requires sufficient data diversity to span the CT→MRI transformation manifold. With 10 training pairs the codebook collapses — generated latents cluster far from the real MRI distribution (Fig. 8a). Flow Matching sidesteps this: MSE on interpolated pairs is well-posed even at 10 samples, and multi-step integration accumulates small, stable corrections rather than forcing a single global mapping.

Raw RL Can Learn Latent Transport but Lacks Strong Priors An interesting observation is that reinforcement learning alone is already capable of learning meaningful latent transport trajectories. As illustrated in Fig. 9, the learned trajectories progressively move from the CT latent distribution toward the MRI latent manifold through adaptive non-linear transitions, demonstrating that sequential decision making is a feasible formulation for cross-modality

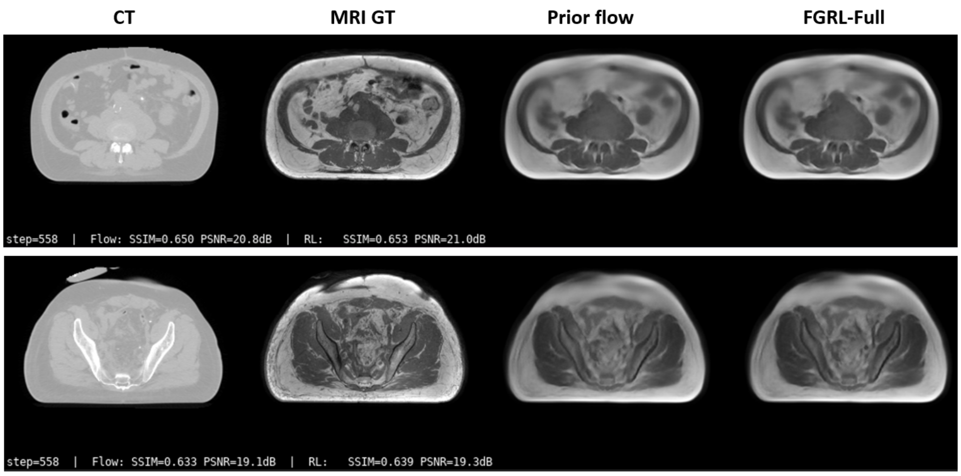


Figure 6: Comparison between the Prior Flow Generator and FGRL-Full. From left to right: input CT image, ground-truth MRI, Prior Flow output, and FGRL-Full output. The quantitative scores shown below each image indicate that FGRL consistently improves upon the Flow Matching baseline, although the visual differences are often subtle.

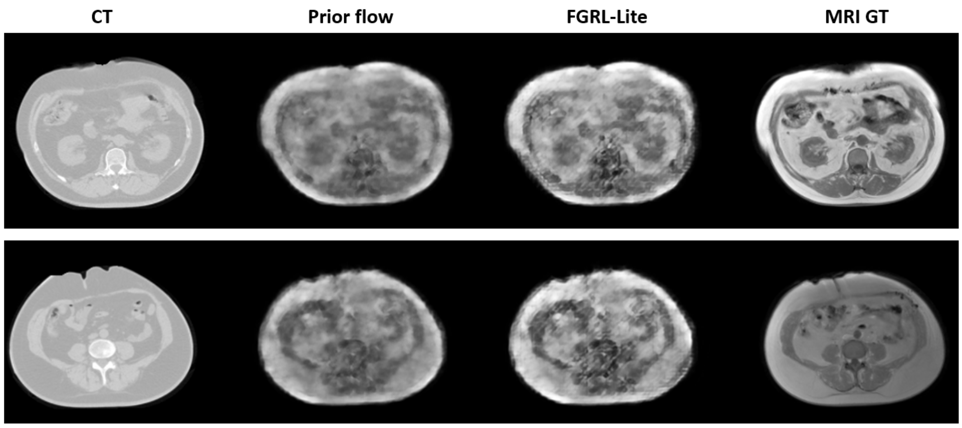


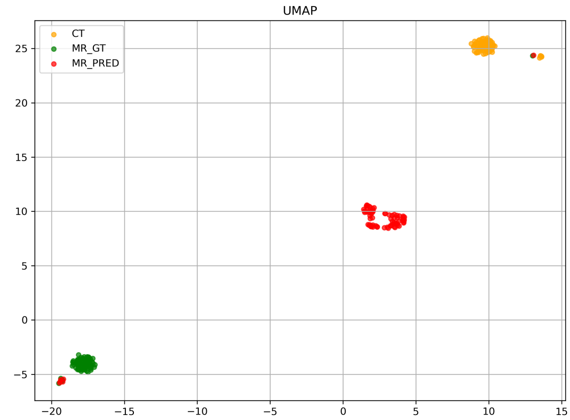
Figure 7: Comparison between Flow Matching and FGRL-Lite. From left to right: input CT image, ground-truth MRI, Prior Flow output, and FGRL-Lite output. The RL policy performs local trajectory corrections that significantly improve anatomical details over the lightweight Flow Matching baseline.

translation.

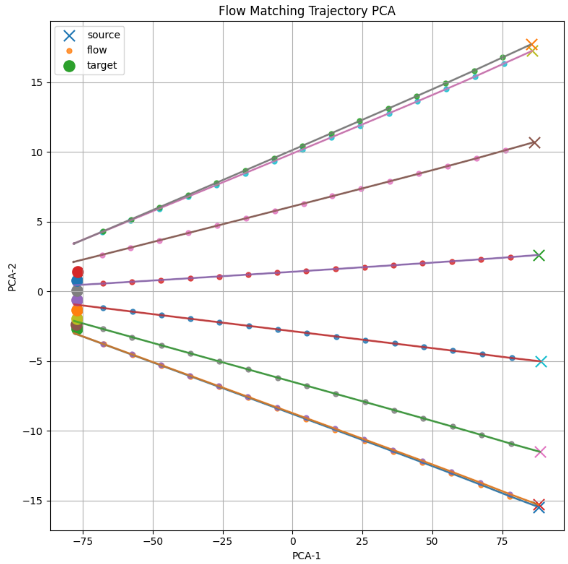
The optimization process is also stable. As shown in Fig. 11, the cumulative reward steadily increases during training, while the trajectory progress toward the target latent distribution improves consistently. Meanwhile, the latent distance between the predicted representation and the target MRI latent gradually decreases, indicating that the policy successfully learns a meaningful transport strategy rather than random exploration.

However, despite learning adaptive latent trajectories, the generated MRI images remain blurry and anatomically inconsistent, as illustrated in Fig. 10. Quantitatively, the raw RL model achieves only moderate reconstruction quality, suggesting that reinforcement learning alone is insufficient to accurately model the complex CT-to-MRI transformation. Without an explicit transport prior, the policy must simultaneously discover both the global transport direction and local refinement strategy from sparse rewards, leading to suboptimal image quality.

These observations motivate the proposed Flow-guided Reinforcement Learning framework, where



(a) UMAP visualization comparing ALDM latent predictions with real MRI latent representations under limited-data training.



(b) PCA visualization of latent transport trajectories. FGRL introduces adaptive non-linear corrections on top of the Flow Matching transport path.

Figure 8: Visualization of the latent transport space. Left: UMAP embedding of ALDM predictions and MRI latent representations. Right: PCA visualization of latent transport trajectories generated by Flow Matching and the proposed Flow-guided Reinforcement Learning framework.

Flow Matching first provides a globally consistent latent transport prior and the RL policy focuses only on adaptive sample-specific trajectory refinement.

7.2 Why RL Can Improve an Already-Good FM

FM trajectories are *globally optimal* (straight-line interpolation during training) but *sample-agnostic* — the same velocity field is applied to every CT input. The RL policy sees the current latent z_t and applies *input-specific* corrections via $a_\perp$ (non-linear manifold correction) and $a_\|$ (adaptive speed modulation per step), providing sample-level trajectory refinement that a fixed velocity field cannot.

7.3 Reward Design Affects SSIM/PSNR Tradeoff

Earlier PPO runs (35, 38) with dense image-space rewards improved SSIM (+0.0038–+0.0041) but degraded PSNR (−0.3 dB), as the policy learned to optimise non-MSE perceptual terms at intermediate steps — causing subtle pixel-level distortions. Run 34 switches to **terminal-only rewards** (SSIM and discriminator at the final step only), removing the per-step perceptual bias. The result: both SSIM (+0.0062) and PSNR (+0.06 dB) improve over FM simultaneously, demonstrating that carefully designed reward timing eliminates the perceived tradeoff.

7.4 Algorithm Choice: Why PPO Over REINFORCE and SAC

PPO vs. REINFORCE. REINFORCE suffers from high gradient variance when the rollout distribution shifts between updates, which becomes pronounced when each episode spans 16 ODE steps with dense rewards. PPO's clipped ratio bounds the effective step size, keeping policy updates conservative even when advantage estimates are noisy. Multiple epochs per rollout (K=4) further improve sample efficiency without the off-policy instability of SAC.

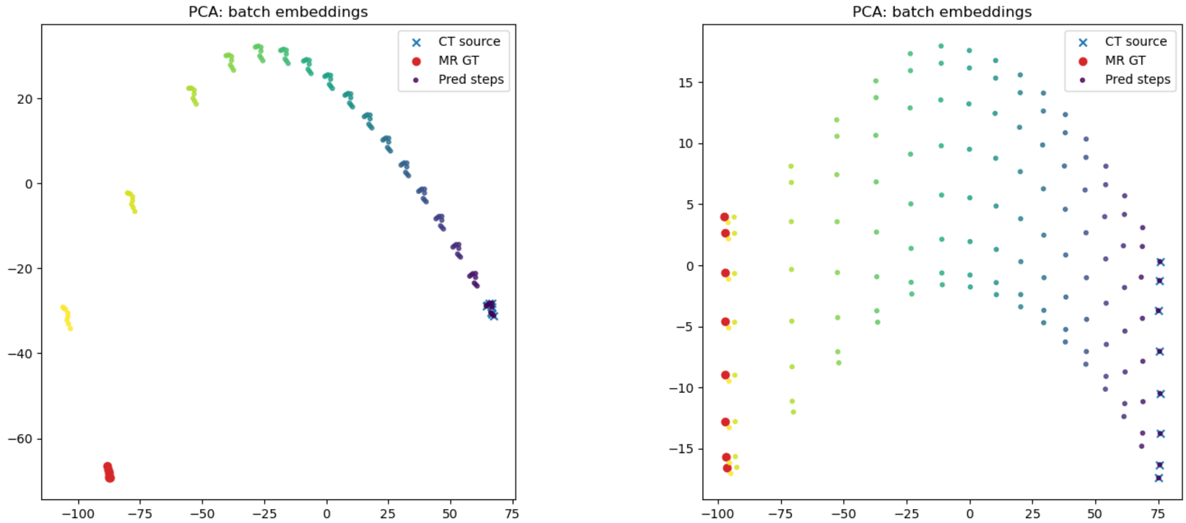


Figure 9: PCA visualization of latent trajectories generated by the raw RL policy under the small-scale training setting. The trajectories progressively move from the CT source distribution toward the MRI target distribution through adaptive non-linear transport paths.

PPO vs. SAC. SAC was initially attempted for its sample efficiency. However, SAC relies on a Q-function trained from a replay buffer. In the post-flow setting, $\text{dist}(z_{\text{flow}}, z_{MR}) \approx 0.09$ — the reward variance across actions falls below 10^{-3}, below the noise floor of the Q-network estimator. The Bellman target collapses to $Q(s, a) \approx 0$ for all actions; the actor receives no gradient. PPO avoids this entirely by operating on-policy: each rollout starts from z_{CT} (dist ≈ 1.09), ensuring non-zero reward variance at every step.

Interleaved ODE compatibility. Each PPO rollout corresponds to one complete 16-step ODE trajectory. The RL correction δv_{RL} is applied at every step, so the policy implicitly learns a trajectory-level strategy rather than a single-step correction. The conservative clip (ε=0.1) is essential: a large policy update at any one ODE step propagates through the remaining steps and can corrupt the globally consistent FM transport direction.

7.5 Limitations

RL gains are modest in both settings (+0.02 REINFORCE, +0.0067 PPO interleaved). A small action scale is required for stability, limiting the per-step correction magnitude. Evaluation covers a single test patient (PPO) and a single scan (REINFORCE), leaving generalization across anatomy and scanner protocols as an open question.

8 Conclusion

We addressed CT-to-MRI synthesis under a fundamental clinical constraint: severely limited paired training data. Our three-stage progression — one-step alignment (ALDM) → multi-step ODE transport (FM) → RL-guided adaptive transport — maps directly to the hypothesis that more steps and adaptivity improve synthesis quality in the data-scarce regime.

Key findings: (1) one-step methods collapse under 10 training pairs; (2) multi-step FM alone surpasses ALDM trained on the full dataset (SSIM 0.79 vs. 0.67 with ∼90× less data); (3) RL consistently improves FM in both REINFORCE (toy) and PPO (full-scale) settings; (4) the critical design choice is *where* RL acts — interleaved ODE intervention outperforms post-flow correction by ∼9× due to richer reward landscapes.

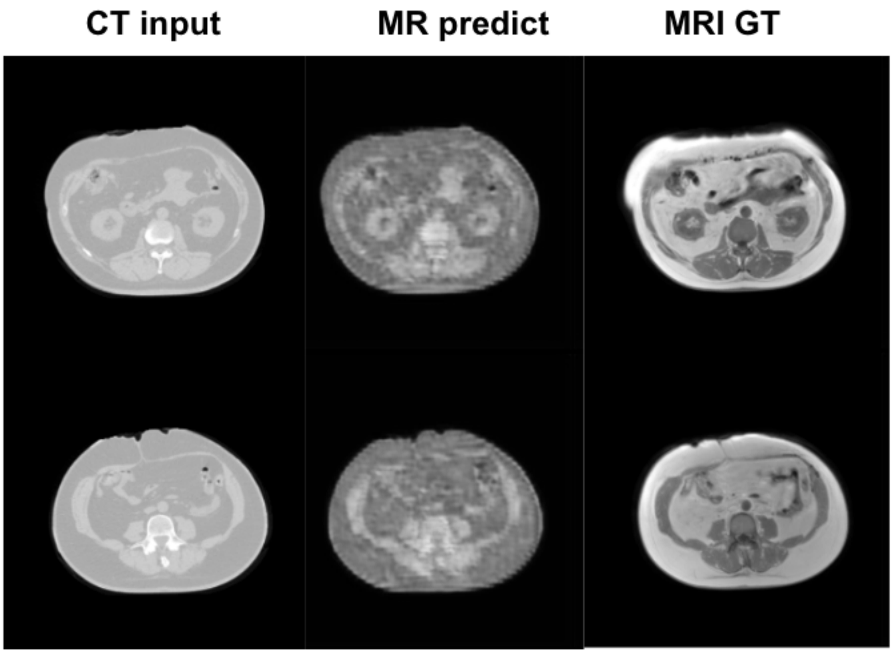


Figure 10: Qualitative CT-to-MRI translation results generated by the raw RL model under the small-scale training setting. Although the model learns progressive latent transport trajectories, the predicted MRI images remain blurry and exhibit noticeable anatomical inconsistencies compared to the ground-truth MRI.

Among RL algorithms, PPO with clipped updates is most suitable for the sequential ODE setting. SAC fails due to Q-collapse in the near-converged latent space, a failure mode specific to off-policy methods when reward variance is near zero.

Future Work

Joint FM+RL training, adaptive action-scale scheduling, evaluation across all three SynthRAD2025 anatomical regions, and replacing the fixed 16-step ODE with an RL-guided adaptive step-size controller.

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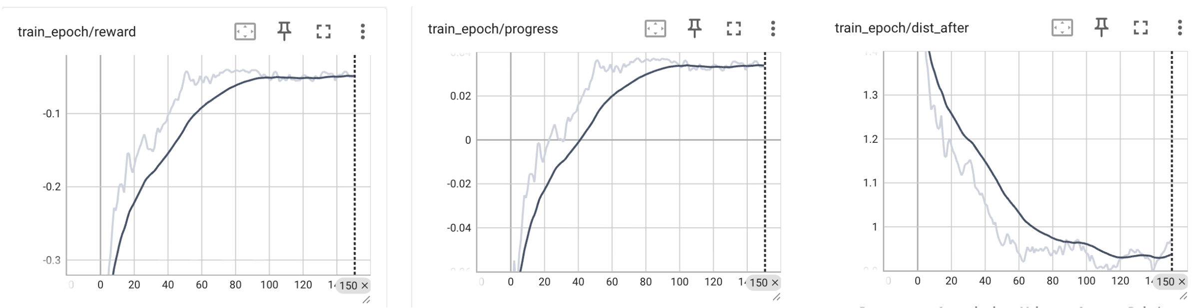

(a) Reward (b) Progress (c) Latent distance

Figure 11: Training dynamics of the raw RL latent transport policy. The plots illustrate the reward evolution, trajectory progress, and latent distance behavior during optimization.